Group Assignment: Analytics on Complex Data

Subject

Presented by
Group Members (3 Names)



Assignment Overview

- This assignment focuses on **applying data analytics techniques** to **complex data**.
- **Complex data** types go **beyond conventional structured datasets (such as csv)** and include various formats, high-dimensional attributes, and intricate relationships. Classification of complex data requires specialized methods tailored to the characteristics of each data type, which differ from traditional data
- **Complex data** refers to non-traditional, large-scale, and heterogeneous data that requires **specialized analytics techniques** for their analysis.
  - Diverse Data Sources, Imbalanced Data
  - High Dimensional Data, Image and Video
  - Text and NLP, Time-Series and Streaming Data
  - Graph and Network Data, Multimodal Data
  - Spatial and Geographic Data
  - Etc.
- **Assignment Objective**
  - Evaluate how your group able to **handling these complex dataset** with case studies and real-world applications
  - Your group will explore, preprocess, analyze, and evaluate a **complex dataset** while demonstrating your findings through a presentation.

Data Collection and Data Sources

Diverse Data Sources

- **Primary Source**
  - Data collected firsthand through direct measurement (or observation)
  - **Examples**:
    - IoT devices, sensors, surveys, laboratory experiments, biometric data.
    - Etc.
- **Secondary Source**
  - Data obtained from external repositories, often requiring extraction, transformation, and integration
  - **Examples**:
    - Web APIs, web scraping, government or business databases, public datasets (e.g., OpenStreetMap, Kaggle).
    - Etc.

. Data warehouse

Characteristics of Complex Data

- **Imbalanced Data**
  - One class has a much **higher number of samples (majority class) than other classes (minority class)**.
  - For example, one class may constitute 99% of the data, while the other accounts for only 1%.
  - As a result, machine learning models tend to learn patterns **favoring the majority class while ignoring the minority class**.
  - **Examples**:
    - **Fraud Detection:** Most credit card transactions are legitimate, while fraudulent transactions represent only a small fraction of the data.
    - **Rare Disease Classification:** Hospital databases contain a large amount of data on common diseases, but rare diseases have very few recorded cases.

- **High Dimensional Data**
  - Complex data often contains a **large number of variables**, making analysis challenging due to the **curse of dimensionality**.
  - **Examples**:
    - **Genomic datasets** (thousands of genetic markers per sample).
    - **Image datasets** (millions of pixel values per image).
    - Sensor networks (multiple sensors generating multi-dimensional streams).

Characteristics of Complex Data

- **Image and Video**
  - Data has **high dimensionality** and requires significant computational power.
  - **Specialized feature extraction** is needed.
  - **Examples**:
    - **Face Recognition**: Used in security systems such as Face ID.
    - **Object Detection in Video**: Used in CCTV security systems or autonomous vehicles.
- **Text and NLP**
  - Data is **unstructured** and comes in **various formats**.
  - The same word can have multiple meanings.
  - **Specialized data pre-processing** is needed.
  - **Examples**:
    - **Sentiment Analysis**: Classifies customer reviews as "positive" or "negative."
    - **Fake News Detection**: Differentiates real news from fake news.

Characteristics of Complex Data

- **Time-Series and Streaming Data**
  - Data changes over time.
  - Must be able to classify in real-time.
  - **Examples:**
    - **ECG Signal Classification:** Predicting heart attack risks from ECG signals.
    - **Real-time Cyber Attack Detection:** Identifying cyber attacks as they happen.

- **Graph and Network Data**
  - Data is structured as a network and cannot be represented in a standard tabular format.
  - Relationships between nodes impact classification outcomes
  - **Examples:**
    - **Terrorist Group Detection from Social Networks.**
    - **Protein Classification Based on Graph Structures.**

Characteristics of Complex Data

- **Multimodal Data**
  - Data originates from multiple sources and comes in different formats.
  - Requires techniques to effectively combine and align different data types.
  - **Examples:**
    - **Disease Diagnosis from Medical Images and Patient Records** (e.g., Combining X-ray and Patient History).
    - **User Behavior Analysis from Audio and Chat Messages.**

- **Spatial and Geographic Data**
  - Data contains **geographic coordinates**, requiring distance and environmental context consideration.
  - Some data points may be **missing or incomplete**.
  - **Specialized data pre-processing** is needed.
  - **Examples:**
    - **Flood Risk Classification from Satellite Images.**
    - **Deforestation Analysis in Forested Areas**

Characteristics of Complex Data

- **Variety**
  - Unclean Structured data (Imbalanced, Noise, …),
  - Unstructured data (Text, Social media posts),
  - Semi-structured data (Logs, Emails),
  - Multi-Media data (Image, Video),
- **Velocity**
  - Changing data
    - Real-time sensor data, stock market feed
  - Fast-moving data streams.
- **High Volume (Big Data)**
  - **…**

Complex Data ? Applications ? (1-2 slides)

Complex Data Exploration (1-2 slides)

Complex Data Pre-Processing (1-2 slides)

Analytic technique (1-2 slides)

- Suggest choosing technique not covered in class

Demo with tool or libraries or coding

Evaluation (1-2 slides)

List of possible application(s)

List of references and/or website(s)

Group Assignment Evaluation Score

- Data Complexity (10%),
- Complex Data Exploration (10%),
- Complex Data Pre-processing (20%),
- Analytics Technique (20%),
- Completeness (20%),
- Presentation + Question (10%)

Possible Subjects

- **Possible Subjects**
  - **Text and Web Analytics**
  - **Opinion Analytics**
  - **Multi-media Analytics**
  - **Graph and Social network Analytics**
  - **Time-series Data Analytics**
  - **Sequential Data Analytics**
  - **High-dimensional Data Analytics**
  - **Stream Data Analytics**
  - **Etc.**

Deadline:

- Subject-Submission & Group-Members: October - 05
- Video, PPT, Report Submission: October - 19